

Aggie Hacks 2025 – FAQ (Frequently Asked Questions)

Q1. Why are the dates in `disaster\_events.csv` from 2010–2015, while the rest of the datasets are from 2023?

This is intentional. The `disaster\_events.csv` file is meant to simulate historical disaster data. It serves as background reference material to help teams understand how cascading disasters unfold. You can use this to train models or infer cause-effect patterns (e.g., flood → outage → hospital overload), but it won't align directly with the real-time 2023 sensor data. You may shift dates for analysis if you clearly document that change in your summary.

Q2. Can we modify or add to the synthetic datasets?

Yes, you're welcome to enrich the data if it improves realism. You may add additional rows to simulate more complex or plausible crisis scenarios. Just be sure to explain any modifications in your one-page summary and how they improve your system's detection, prediction, or decision-making capabilities.

Q3. Why are some sensor values between 0 and 100? Are these real units?

No — these are normalized synthetic scores. Think of them as intensity levels: 0 = normal, 100 = extreme. This applies to all sensor types (flood, seismic, humidity, temperature). They're designed to mimic a risk scoring system rather than real-world units like °C or meters.

Q4. Why do sensor locations seem to move? Isn't each sensor fixed?

Yes, in real life sensors would be fixed. But we introduced slight shifts in GPS coordinates to simulate GPS drift, signal noise, or sensor misclassification - common issues during crises. Your system should be robust enough to reason through noisy or imperfect geospatial data.

Q5. Why isn't the system detecting any fake tweets, is there any problem with the data?

No, The detection logic depends on your implementation. If you're using contradiction logic, the system should flag misinformation. If you're not seeing any flagged tweets, consider enriching your dataset or revisiting your logic.

Q6. Can we test tweets manually to see if they're fake or real?

Yes! You are welcome to build that functionality into your solution if you'd like! One suggestion is to allow users to input a tweet along with a timestamp and location, and then compare it against current sensor and disaster data to assess if it's misinformation.

Q7. Can we use zone names like 'Zone A', 'Zone B' instead of lat/lon-based zones?

We've intentionally provided a mix of formats across datasets. Some files use structured zone names (like Zone A, Zone B), while others rely on latitude/longitude coordinates. You're free to work with either - or even build a translation layer that maps lat/lon to zone labels. What matters most is consistency within your solution. If you choose to work with lat/lon zones, rounding to one decimal place is a helpful way to group readings geographically.

Q8. Can we assume cascading logic (flood → power outage → hospital overload) even if it's not clearly visible in the data?

Yes, and that's part of the challenge. You are encouraged to simulate and model cascading chains using reasonable assumptions or create triggers from combined datasets. Just document your approach.

Q9. Why aren't there more obvious disasters in the dataset?

This is a deliberate part of the simulation. You are expected to uncover patterns, simulate events, and demonstrate how your AI would act if the city was under stress — even if everything isn't handed to you explicitly.

Q10. Are we allowed to simulate viral tweets or press releases?

Yes! If you want to simulate misinformation going viral, feel free to enrich the `social\_media\_stream.csv` with more entries, as long as it's clearly labeled and justified.

Q11. Should we include everything in our presentation?

Yes, You'll submit a final deck (no more than 9 Slides) and a one-page summary. You can show your live demo during the final presentation round (Optional). Be sure to document any assumptions, external techniques, or enrichments you made to the data.

Q12. How to interpret energy_kwh vs. real blackouts

Not directly. The energy_kwh reflects the electricity usage of building level at specific timestamps. A sudden drop or spike might indicate a blackout or power surge. You can create rules for example: 3 buildings in Zone A show 0 Kwh for 10+ mins which indicates a blackout. If you're creating a rule, please document in the summary.

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Q13. What does shifting the dates mean? (Follow up from question 1)

This means modifying the timestamps in the disaster_events.csv file to match the year 2023, so they align with the sensor and real-time data.

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Q14. The dataset doesn't match with other dataset - how are we supposed to combine into one?

This is expected. Each Dataset serves a different purpose and should be used accordingly. You don't need a perfect merge. You need smart logic to reason across imperfect, noisy data